Mo Tu We Th Fr Sa Su

Memo No.

Date / /

Week 8 Day 3

Manual Testing

- **Test Plan**

It is a doc that will explain what will be test, who will do what, how that would be tested, when it will be tested, and what resource will it requires.

- **Time Estimation**

i) Expert Judgment

Experienced testers estimate the required time based on their knowledge and previous projects.

ii) Work Breakdown Structure

It will break the entire testing project into smaller activities and estimate the time required for each activity.

iii) Three Point Estimation

It uses three estimation

a) Optimistic (O): Best-case time

b) Most Likely (M): Normal expected ti

c) Pessimistic (P): Worst-case time

Memo No. _____
Date / /

$$E = \frac{O + 4M + P}{6}$$

- Test Case

It is a set of action performed to verify perticular funcanility or feature of an application

- Effective Ways to write test casees
  - Clarity (Write of a reader)
  - One test case and one clear purpose
  - Cover the full scope - but don't go beyond it
  - Make it traceable
  - Keep it independent if possible.

- Shift Left Testing

In this testing approach the testing activities are started early in the starting phase of SDLC help to find defect in early stage of development.

- WCAG (Web Content Accessibility Guidelines)

It is a standards for creating a web

Mo Tu We Th Fr Sa Su

Memo No.
Date / /

content accessible to people with disabil.
It have four principles: Perseivable, operable, Understandable, Robust. It is divided into 3 level, A, AA, AAA.

- Type of API Architectures
  i) Rest API

Reperesentational state Transfer is a HTTP method used to enable communication between server and client using JSON (Java Object Notation)

  ii) Graph QL

It is a query language that make request for the exact data that are required

  iii) Web socket

Web Socket used For Real time communicat with each other in real time using persistent connection.

- JWT (JSON WEB TOKEN)

It is a secure and compact method of transmitting information between two parties

Date / /

- OAuth

It allows secure third-party access without sharing user credentials

- Idempotency

A same input should always give same output.

- API Versioning

Design API in a such way that updating does not break the existing client app.

- ACID Properities

These properities ensure that the database transaction are reliable and accurate

i) Atomicity

Either complete or nothing either

ii) Consistency

Before and after of an transaction that states must be same

iii) Isolation

Multiple transaction run simultaneously with out affecting other

Mo Tu We Th Fr Sa Su

Memo No.

Date / /

iv) Durability

Once commited the state remains same no changing afterwards.

- OWASP (Open Web App Security Project)

It yearly ~~prom~~ publish list of Top 10 web vencribilities

i) Broken Access point
ii) Cryptographic failure
iii) Injection
iv) Insecure Design
v) ~~Outdated~~ Outdated Packages
vi) Security Misconfiguration
vii) Identification & Authorization failure
viii) Software & Data Integrity failure
ix) logging & Monitering failure
x) ~~Cross Site~~ Server Side Request forge

- SQL Injection

Attecker try to run malicious queries on a database

Mo Tu We Th Fr Sa Su

Memo No. _____

Date / /

- Cross Site Scripting

Attacker inject malicious exe. code into the ~~ob~~ website affecting other users.